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Device configured to be mounted on an oil tank of an aircraft turbomachine, oil supply assembly and associated method of use

Abstract

An adapter is configured to be disposed on the inlet of an oil tank of an aircraft engine. The adapter includes a plug between the adapter and the exterior, and a flapper between the adapter and the oil tank. An adapter cavity located between them is set to negative pressure during flights. When the aircraft is back on the ground, this negative pressure has to be counteracted to open the plug, which can make the opening operation difficult. The adapter includes a lever that counteracts the negative pressure in the cavity of the adapter to allow easy opening of the plug with limited force.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a device configured to be mounted on an oil tank of an aircraft turbomachine. The present invention also relates to an oil supply assembly comprising at least one such device. The present invention further relates to a method of using such a device.

TECHNICAL BACKGROUND

(2) Known aircraft engines comprise many components that require lubrication in order to operate efficiently. As such, the engines comprise oil tanks containing oil for lubricating engine components, such as gearboxes. EP 3 399 164 describes such an oil tank.

(3) In order to supply oil to an engine, a person generally pours oil into the engine's oil tank, via an adapter disposed on the inlet of the oil tank. In order to ensure good fluid isolation from the oil tank, the adapter comprises two seals, a primary seal called the "plug" between the adapter and the outside of the adapter, and a secondary seal called the flapper between the adapter and the oil tank. An adapter cavity is located between these two seals.

(4) During an aircraft flight, the adapter cavity is evacuated. When the aircraft returns to the ground, this vacuum must be counteracted to open the plug, which can make the opening operation difficult.

SUMMARY OF THE INVENTION

(5) One objective of the invention is to assist in opening the plug of an aircraft turbomachinery oil tank. In particular, the invention proposes to counteract the negative pressure in the adapter cavity, and to allow easy plug opening with limited force.

(6) To this end, the object of the invention is a device configured to be mounted on an oil tank of an aircraft turbomachine, comprising: a channel located between a device inlet and a device outlet, the device outlet being intended to be connected to the oil tank, a plug movably mounted on the device inlet and configured to take: a venting position in which fluid can enter the channel through the device inlet, and a sealing position in which the plug completely seals the device inlet; the device inlet comprising a projection and in that the device comprises a lever connected to the plug; the lever comprising a bearing elbow capable of bearing on the projection and pivoting about the projection between a projection contact position and a lifting position; the plug being in the sealing position when the lever is in its projection contact position, the plug being in the venting position when the lever is in its lifting position; and the lever moving the plug from its sealing position to its venting position as it pivots from the projection contact position to the lifting position.

(7) According to an embodiment, the lever is movable between a rest position and the projection contact position, the lever being away from the projection when it is in the rest position, the plug being in the sealing position when the lever is in the rest position.

(8) According to an embodiment, the projection comprises an orifice, and the plug comprises a closure pin able to enter the orifice so as to block the plug in its sealing position; when the lever is in its rest position, the closure pin is inserted into the orifice; when the lever is in its projection contact position, the closure pin being away from the orifice; and the rotation of the lever from its rest position to its projection contact position being configured to withdraw the closure pin from

the orifice.

(9) In one embodiment, the lever is secured to the closure pin by a lever pivot, the lever being pivotable about the lever pivot between its rest and projection contact positions.

(10) According to one embodiment, the lever comprises a short arm extending between the bearing elbow and the lever pivot, the short arm having a first length, and a long arm forming a handle and extending over a second length from the bearing elbow, the second length being at least three times greater, advantageously five times greater, than the first length between the bearing elbow and the lever pivot.

(11) According to one embodiment, the interior of the device is configured to be under negative pressure when the plug is in its sealing position.

(12) According to one embodiment, the device comprises a filter disposed in the channel between the plug and the device outlet.

(13) The invention also relates to an oil supply assembly comprising an oil tank having a tank inlet and at least one device as described above, the device outlet being connected to the tank inlet.

(14) The invention also relates to a turbomachine comprising an oil supply assembly as described above.

(15) The invention also relates to an aircraft comprising at least one turbomachine as described above.

(16) The invention also relates to a method of using a device as described above, comprising the following steps: sealing the device inlet with the plug in its sealing position; causing the lever to bear against the projection; pivoting the lever about the projection between the projection contact position and the lifting position, the lever moving the plug from its sealing position to its venting position during this pivoting; and fluid entering the device through the device inlet.

(17) The advantages mentioned for the device apply mutatis mutandis to the method.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) Other features and advantages of the invention will become apparent from the following detailed description, for the understanding of which reference is made to the appended figures, of which:

(2) FIG. 1 is a schematic view of an oil tank inlet with a device according to the invention mounted on the tank inlet,

(3) FIG. 2 is a schematic cross-sectional view of a device according to the invention, with the plug in the sealing position and the lever in the rest position,

(4) FIG. 3 is a schematic cross-sectional view of a device according to the invention, with the plug in the sealing position and the lever in an intermediate position between the rest position and the projection contact position,

(5) FIG. 4 is a schematic cross-sectional view of a device according to the invention, with the plug in the sealing position and the lever in the projection contact position, and

(6) FIG. 5 is a schematic cross-sectional view of a device according to the invention, with the plug in the venting position and the lever in the lifting position.

EMBODIMENTS OF THE INVENTION

(7) The present invention is described with particular embodiments and references to figures, but the invention is not limited by them. The drawings or figures described are schematic only and are not limiting. Furthermore, the functions described may be performed by structures other than those described herein.

(8) In the context of the present document, the terms “first” and “second” are used solely to differentiate the various elements and do not imply any order between these elements.

(9) In the figures, identical or similar elements may bear the same references.

(10) FIG. 1 shows an oil supply assembly **10**. The oil supply assembly **10** comprises an oil tank **12** and at least one device **14** mounted on the oil tank **12**.

(11) The oil tank **12** has a tank inlet **18**. The device **14** is connected to the tank inlet **18**.

(12) The device **14** according to the invention is in particular an adapter mounted on an oil tank. In particular, the adapter is configured to guide oil from an external container that contains oil to the oil tank when pouring the oil.

(13) The supply assembly **10** belongs, for example, to a turbomachine. Such a turbomachine is typically mounted on an aircraft, for example a civil or military aircraft.

(14) FIG. 2 illustrates the closed device **14**. FIGS. 2 to 5 show successive stages of opening device **14**.

(15) The device **14** comprises a device inlet **22**, a device outlet **24**, and a channel **26** located between the device inlet **22** and the device outlet **24**. The device outlet **24** is connected to the tank inlet **18**.

(16) The device inlet **22** is circular in shape, for example. According to one embodiment, it has an area of between 5 cm.sup.2 and 80 cm.sup.2, and advantageously substantially equal to 20 cm.sup.2.

(17) The device inlet **22** comprises a projection **30** from the channel **26**. According to the embodiment shown in FIGS. 2 to 5, the projection **30** extends substantially parallel to the channel **26** and extends in an opposite way from the device outlet **24** with respect to the device inlet **22**.

(18) Advantageously, the projection **30** comprises a hole **32** configured to receive a closure pin **34**, as will be described below.

(19) Advantageously, the device outlet **24** comprises a sealing membrane **36** configured to at least partially seal the channel **26**. The sealing membrane **36** is intended to seal the device **14** when an undesirable leak occurs around the plug **38**. In this case, the sealing membrane **36** is pressed against the outlet of the device **24** by the internal pressure of the oil tank **12**.

(20) The device **14** also comprises a plug **38**. The plug **38** is movably mounted on the device inlet **22**. The plug **38** is configured to assume a venting position **101** (illustrated in FIG. 5) in which fluid (in particular air) is able to enter the channel **26** through the device inlet **22**, and a sealing position **100** (illustrated in FIGS. 2 to 4) in which the plug **38** completely seals the device inlet **22**.

(21) Advantageously, the plug **38** is also able to move to an open position which is beyond its venting position. In the open position the plug **38** is completely away from the device inlet **22**. In one embodiment, the plug **38** in its open position abuts an edge surrounding the device inlet **22** so that the plug **38** is held in the open position without user intervention. The open position of the plug **38** allows the user to pour oil through the entire cross-section of the device inlet **22**.

(22) According to the embodiment shown in the figures, the plug **38** comprises a closure pin **34**. The closure pin **34** is displaceable. According to the embodiment shown in the figures, the closure axis **34** is movable substantially perpendicular to the channel **26**. The closure pin **34** is able to enter the orifice **32**, as shown in FIGS. 2 and 3, so as to block the plug **38** in its sealing position **100**.

(23) According to one embodiment, the closure pin **34** comprises a lever pivot **40**.

(24) Advantageously, the plug **38** comprises a seal **42** intended to come into contact with the channel **26** when the plug **38** is in its sealing position **100**. The seal **42** allows the plug **38** to better isolate the channel **26** when the plug **38** is in its sealing position **100**.

(25) When the plug **38** is in its sealing position **100**, the interior of the device **14** is subjected to negative pressure during flight. During ascent to altitude and during the flight phase at maximum altitude, the interior of device **14** will be at excess pressure relative to atmospheric pressure. During descent, the pressure inside device **14** will remain at the minimum pressure reached at altitude. On return to the ground, the pressure inside the device **14** may in some cases be lower than atmospheric pressure, which generates the negative pressure that must be overcome when the plug **38** is opened. According to one embodiment, the inside of the device has a pressure of less than

0.95 bar when the plug **38** is in its sealing position **100**, or less than 0.92 bar, or even less than 0.9 bar.

(26) Advantageously, the device **14** comprises a filter **44** disposed in the channel **26** between the plug **38** and the device outlet **24**. The filter **44** is capable of retaining particles larger than 8 mm, advantageously larger than 6 mm, in order to prevent them from passing through the channel **26**.

(27) According to the embodiment shown in the figures, the filter **44** has a substantially hemispherical shape with its dome extending away from the plug **38**. This shape is particularly suitable for preventing particles coming from the device inlet **22** from reaching the device outlet **24**.

(28) The device **14** also comprises a lever **48** connected to the plug **38**. According to the embodiment shown in the figures, the lever **48** is connected to the plug **38** by the lever pivot **40**.

(29) FIG. 2 shows lever **48** in its rest position **200**; FIG. 4 shows lever **48** in its projection contact position **201**; and FIG. 5 shows lever **48** in its lifting position **202**.

(30) The lever **48** comprises a bearing elbow **52** capable of bearing on the projection **30**. The lever **48** comprises a short arm **54** extending between the bearing elbow **52** and the lever pivot **40**, and having a first length **56**.

(31) The lever **48** has a free lever end **58** opposite the bearing elbow **52**. It is intended for a user to exert a force thereon to open the device **14**. The lever **48** comprises a long arm **60** extending between the bearing elbow **52** and the lever end **58**, and having a second length **62**. This second length **62** is at least three times, preferably five times, greater than the first length **56** of the short arm **54**. In other words, the lever **48** has a mechanical advantage greater than 3, preferably greater than 5.

(32) With reference to FIGS. 2 to 4, the lever **48** is able to pivot about the lever pivot **40** between a rest position **200** (shown in FIG. 2) and a projection contact position **201** (shown in FIG. 4). The lever **48** is able to be a plurality of intermediate positions (one of which is illustrated in FIG. 3) between its rest position **200** and its projection contact position **201**. When the lever **48** is in the rest position **200**, the lever **48**, in particular the bearing elbow **52**, is away from the projection **30**. When lever **48** is in the rest position **200**, the plug **38** is in the sealing position **100**. When lever **48** is in its rest position **200**, the closure pin **34** is inserted into the orifice **32**. Thus, a potential movement of the plug **38** from its sealing position **100** to its venting position **101** is prevented by the cooperation of the closure pin **34** with the orifice **32**.

(33) Advantageously, device **14** comprises a spring (not shown in the figures) configured to hold lever **48** in its rest position when the pivoting force exerted on lever **48** is below a threshold. The spring is located, for example, between the plug **38** and the closure pin **34**.

(34) Rotation of the lever **48** from its rest position **200** to its projection contact position **201** removes the closure pin **34** progressively from the orifice **32**.

(35) When the lever **48** is in the projection contact position **201**, the lever **48** is in contact with the projection **30**. According to the embodiment shown in the figures, when the lever **48** is in the projection contact position **201**, the bearing elbow **52** is in contact with the projection **30**.

(36) When the lever **48** is in the projection contact position **201**, the plug **38** is in the sealing position **100**. Thus, during the movement of lever **48** between its rest position **200** and its projection contact position **201**, the plug **38** always remains in the sealing position **100**.

(37) When the lever **48** is in its projection contact position **201**, the closure axis **34** is away from the orifice **32**.

(38) With reference to FIGS. 4 and 5, after the lever **48** contacts the protrusion **30**, the lever **48** is able to pivot about the protrusion **30** between the protrusion contact position **201** (shown in FIG. 4) and a lifting position **202** (shown in FIG. 5).

(39) When the lever **48** is in its lifting position **202**, the plug **38** is in the venting position **101**. As the lever **48** pivots from the protrusion contact position **201** to the lifting position **202**, the lever **48** moves the plug **38** from its sealing position **100** to its venting position **101**.

(40) The method of using the device **14** will now be briefly described.

(41) Firstly, as illustrated in FIG. 2, the plug **38** is in its sealing position **100** and the lever **48** is in the rest position **200**. The closure pin **34** is inserted into the orifice **32**. The device inlet **22** is sealed by the plug **38** so that no fluid can enter or leave the channel **26**.

(42) When a user wishes to change the plug **38** to its venting position **101**, he pivots the lever **48** about the lever pivot **40**. The lever **48** passes through the intermediate positions before arriving at its projection contact position **201**. During this pivoting of the lever **48** between its rest position **200** and its projection contact position **201**, the closure pin **34** gradually moves out of the orifice **32**.

(43) The resistive force during pivoting of the lever **40** between its rest position **200** and its projection contact position **201** is principally the friction between the lever **48** and the closure pin **34** at the lever pivot position **40**, and the friction between the closure pin **34** and the orifice **32**.

(44) According to the embodiment where the device **14** comprises the spring as described above, the resisting force during pivoting of the lever **40** from its rest position **200** to its projection contact position **201** further comprises the restoring force on the lever **48** towards its rest position.

(45) Once the lever **48** reaches its projection contact position **201**, the bearing elbow **52** bears against the projection **30**. To continue pivoting the lever **48** from the projection contact position **201** to the lifting position **202**, the user exerts a force on the lever **48**, advantageously on the lever end **58**, to counteract the resistance force. The resisting force during pivoting of the lever **48** between its projection contact position **201** and its lifting position **202** is primarily the vacuum inside the device **14** relative to the outside of the device **14**. The force associated with this negative pressure is greater than the friction on lever **48** during pivoting of lever **40** between its rest position **200** and its projection contact position **201**. With the use of the lever **48**, the force required to open plug **38** is much smaller compared to when opening the plug **38** directly without using the lever **48**.

(46) As the lever **40** pivots from its projection contact position **201** to its lifting position **202**, the lever **48** progressively moves the plug **38** from its sealing position **100** to its venting position **101**.

(47) Once the plug reaches its venting position **101**, fluid (in particular air) is able to enter the device **14** through the device inlet **22**.

(48) Advantageously, to pour oil through the channel **26**, the user continues to control the lever **48** from its venting position to its open position, for example by pulling the lever **48**. According to one embodiment, the lever **48** is in contact with the plug **38** during at least part of the pivoting of the lever **48** between its venting position and its open position.

(49) In other words, the invention relates to an adapter disposed on the inlet of an aircraft engine oil tank **12**. The adapter comprises a plug **38** between the adapter and the exterior, and a flapper between the adapter and the oil tank. An adapter cavity **26**, located between them, is set to negative pressure during flights. When the aircraft is back on the ground, this negative pressure has to be counteracted to open the plug, which can make the opening operation difficult. The invention proposes, by means of a lever **48**, to counteract the negative pressure in the cavity of the adapter, and to allow easy plug opening with limited force.

(50) The present invention has been described in relation to specific embodiments, which are purely illustrative and should not be considered limiting. Generally speaking, the present invention is not limited to the examples illustrated and/or described above. The use of the verbs “comprise”, “include”, or any other variant, as well as their conjugations, can in no way exclude the presence of elements other than those mentioned. The use of the indefinite article “a”, or of the definite article “the”, to introduce an element does not exclude the presence of a plurality of these elements. The reference numbers in the claims do not limit their scope.

Claims

1. A device configured to be mounted on an oil tank of an aircraft turbomachine, the device comprising: a channel located between a device inlet and a device outlet, the device outlet being configured to be connected to the oil tank, a plug movably mounted on the device inlet and configured to take: a venting position in which fluid can enter the channel through the device inlet, and a sealing position in which the plug completely seals the device inlet; wherein the device inlet comprises a projection and in that the device comprises a lever connected to the plug; the lever comprising a bearing elbow configured to bear on the projection and to pivot about the projection between a projection contact position and a lifting position; the plug being in the sealing position when the lever is in the projection contact position, the plug being in the venting position when the lever is in the lifting position; and the lever moving the plug from the sealing position to the venting position as the lever pivots from the projection contact position to the lifting position.
 2. The device according to claim 1, wherein the lever is movable between a rest position and the projection contact position, the lever being away from the projection when the lever is in the rest position, the plug being in the sealing position when the lever is in the rest position.
 3. The device according to claim 2, wherein the projection comprises an orifice, and the plug comprises a closure pin configured to enter the orifice so as to block the plug in the sealing position; when the lever is in its rest position, the closure pin is inserted into the orifice; when the lever is in its projection contact position, the closure pin being away from the orifice; and rotation of the lever from its rest position to its projection contact position being configured to withdraw the closure pin from the orifice.
 4. The device according to claim 3, wherein the lever is secured to the closure pin by a lever pivot, the lever being pivotable about the lever pivot between rest position and the projection contact position.
 5. The device according to claim 4, wherein the lever comprises a short arm extending between the bearing elbow and the lever pivot, the short arm having a first length, and a long arm forming a handle and extending over a second length from the bearing elbow, the second length being at least three times greater than the first length between the bearing elbow and the lever pivot.
 6. The device according to claim 5, wherein the second length is at least five times greater than the first length between the bearing elbow and the lever pivot.
 7. The device according to claim 1, wherein an interior of the device is configured to be put under negative pressure when the plug is in its sealing position.
 8. The device according to claim 1, further comprising a filter disposed in the channel between the plug and the device outlet.
 9. An oil supply assembly, comprising an oil tank having a tank inlet and at least one device according to claim 1, the device outlet being connected to the tank inlet.
 10. A turbomachine comprising the oil supply assembly according to claim 9.
 11. An aircraft comprising at least one turbomachine according to claim 10.
 12. A method of using the device according to claim 1, the method comprising the following steps: sealing the device inlet with the plug in the sealing position; causing the lever to bear against the projection; and pivoting the lever about the projection between the projection contact position and the lifting position, the lever moving the plug from the sealing position to the venting position during the pivoting; wherein fluid enters the device through the device inlet.
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